

Introduction

We are analyzing a loan approval dataset that uses independent variables such as age, gender, annual income, and credit score, showing whether an applicant was approved for the loan. It uses several different determiners that allow us to understand why an applicant may have been approved or denied a loan. Of these variables, we will be analyzing the correlation of marriage status on loan approval, if it exists. In 1974 the Equal Credit Opportunity Act was enacted, which prohibited creditors from denying an applicant based on marital status, among other variables (*The Equal Credit Opportunity Act*). So, if there is a correlation, we will expand the research to additional variables such as credit score and loan amount in addition to marriage status to ensure that there is not illegal discrimination based on the variable.

Banks and lenders may be able to use this data to understand applicants and how they can tailor their own application process to effectively lend money safely. They can also use this data to find an illegal, underlying bias that may not be noticeable in day-to-day operations. Applicants may be able to use this data to understand if they would be approved for a loan or not, giving them a deeper understanding of why people are approved or denied a loan.

Problem Statement

Given the variety of characteristics we are given in this data set, there are a plethora of questions we can ask and assumptions we can make. The relationship between marital status and one's ability to get a loan is one question that we want to analyze. On paper, there are three options for marital status, single, married, and divorced. These three options encompass the entirety of the population of people applying for loans, so, does this play a role in the approval or denial of a loan? We intend to see if any one of these selections has a higher or lower rate of being approved with reasonable confidence.

Additionally, if we look deeper and analyze how the loan amount and marital status affect approval, applicants can see how levying the amount they are asking for will give them greater chances of approval and predict if they will approve or denied for their loan amount. Another variable we will look at is credit score in conjunction with marriage status, it is already known that a high credit score is helpful in the approval process of loans; so, it can be a key indicator to see if there is discrimination based on our main variable.

Data

This project uses data from [Kaggle](#) uploaded by Amn Amine in 2026 and it is updated monthly. The original data contained 1000 rows and 11 columns. We plan to randomly select a random sample of 250 rows and keep all 11 columns to create the most variability. The data has a few limitations and assumptions to note:

- Due to this data being synthetic, the results may not fully apply to real world situations. According to the creator of the dataset “the features were carefully designed to reflect

typical distributions found in financial datasets” (*Amine*). However, real world outcomes are likely to show greater variability than what is represented in the data.

- The dataset creator only has applicants aged 21 to 64, so results may not be representative for teenagers or elderly people.
- There is an oversimplification of the data in this set. Other factors affect loan approval, including location, loan purpose, and non-financial assets/collateral.
- We assume that the data provided is accurate, unbiased, and represents a large population of loan applicants.
- We assume the variables were measured correctly and free of intentional manipulation.

References

- Amine, Amn. "Loan Approval Dataset." *Kaggle*, 11 Jan. 2026, www.kaggle.com/datasets/amineipad/loan-approval-dataset/data.
- "SUBCHAPTER IV—EQUAL CREDIT OPPORTUNITY." *United States Code, 2011 Edition*, U.S. Government Publishing Office, 2011, www.govinfo.gov/content/pkg/USCODE-2011-title15/html/USCODE-2011-title15-chap41-subchapIV.htm.
- "The Equal Credit Opportunity Act." *Civil Rights Division*, U.S. Department of Justice, 2 Jan. 2025, www.justice.gov/crt/equal-credit-opportunity-act-3.